

RIPHAH INTERNATIONAL UNIVERSITY



**Faculty of Computing
FINAL YEAR PROJECT PROPOSAL & PLAN**

**Digital FIR: A Blockchain-Based Secure First Information Report
Management System**

Project Team

Full Name of Student	SAP Id	Program	Contact Number	Email Address
Sharaiz Ahmed	57288	BSCS	0334-4403444	57288@students.riphah.edu.pk
Abdul Moiz	54482	BSCS	0334-6660574	54482@students.riphah.edu.pk
Basit Ali	54596	BSCS	0311-1559530	54596@students.riphah.edu.pk

Tabassum Javed
(Designation: Senior Lecturer)

Table of Contents

Change Record:	5
Opportunity & Stakeholders:	6
Literature Review of Existing Systems:	6
Market Survey:	8
Research Gap Identified from Existing Work	8
Problem Statement:	9
Proposed Solution:	10
Proposed FIR and Integrity Verification Workflow	10
Step 1 - Complaint Submission.....	10
Step 2 - Initial Storage	10
Step 3 - FIR Review.....	10
Step 4 - FIR Creation	10
Step 5 - Officer Assignment	11
Step 6 - Investigation	11
Step 7 - Verification.....	11
Step 8 - SHA-256 Hash Generation.....	11
Step 9 - Blockchain Transaction	11
Step 10 - Off-Chain Storage	11
Step 11 - Later Integrity Check.....	11
Step 12 - Hash Comparison	11
Step 13 - Verification Result.....	11
Architecture Diagram	11
Functional Requirements:	12
Non-Functional Requirements:	14
Security Requirements:	15
Scope of the Project:	16
Module 1 - Authentication and Role Management.....	16
Module 2 - Complaint and FIR Management	16
Module 3 - Investigation and Officer Assignment	16
Module 4 - Evidence Management and SHA-256 Integrity Verification.....	17
Module 5 - Hyperledger Fabric and Audit Trail	17
Module 6 - Dashboard, Search, Notifications and Administration.....	17
List of Faculty proposed changes:	18
Project Plan	19
Roles & Responsibility Matrix:	20
Approval	21

Digital FIR: A Blockchain-Based Secure First Information Report Management System

Change Record

Author(s)	Version	Date	Notes	Supervisor's Signature
Sharaiz Ahmed, Abdul Moiz, Basit Ali	1.0	15 September 2026	Original Draft	
Sharaiz Ahmed, Abdul Moiz, Basit Ali	1.1	29 September 2026	Changes Based on Feedback From Faculty	
Sharaiz Ahmed, Abdul Moiz, Basit Ali	1.2	29 September 2026	Added Project Plan and Roles & Responsibility Matrix	
Sharaiz Ahmed, Abdul Moiz, Basit Ali	1.3	30 September 2026	Applied Literature Review, Existing System Study, Domain Study and Workflow Findings	
Sharaiz Ahmed, Abdul Moiz, Basit Ali	1.4	30 September 2026	Added citations, references and PoliceChain	

Project Proposal

Project Title: Digital FIR: A Blockchain-Based Secure First Information Report Management System

Opportunity & Stakeholders:

Digital FIR is an academic prototype for secure digital management of FIR records and the related case workflow. An FIR passes through different stages and involves different users, including citizens or complainants, police registration staff, investigating officers, Station House Officers, supervisory officers, system administrators and authorized audit personnel. The opportunity is to provide a structured digital workflow where complaints can be submitted, FIRs can be created and reviewed, investigation activities can be recorded, evidence can be registered and case status can be tracked. The project also provides an integrity and traceability layer for important records and documents. The system will use role-based access so that each user can only perform the actions allowed for their role. The project is not intended to replace existing provincial police systems. Instead, it will demonstrate how a permissioned blockchain can be used as an additional integrity and audit layer for a digital FIR workflow.

Literature Review of Existing Systems:

Digital FIR and police record management systems are already being used or studied in different countries. In Pakistan, police departments have introduced digital systems for complaints, FIRs and police records to reduce paperwork and make it easier to store, access and track information. For example, Punjab Police has systems for complaint management and police station records, while Sindh Police and Khyber Pakhtunkhwa Police also have digital record and complaint systems. These systems mainly use web applications and centralized databases. From our study, we found that digital FIR management is already possible but using blockchain as an additional layer for data integrity and auditing is a separate area of research.

We also studied different blockchain-based FIR and police record systems. Hassija et al. (2021) proposed a consortium blockchain system for FIR registration and tracking in the Indian Police Department. In this system, police stations share a blockchain network and citizens can register FIRs through an application. Khan et al. (2020) proposed Smart FIR, which uses Ethereum and smart contracts to help protect e-FIR data stored in a police database. These studies show how blockchain can be used to improve the security, transparency and tracking of FIR records.

Another system we studied is PoliceChain by Mukherjee and Halder (2020). It is a blockchain-based smart policing system that allows citizens, police officers, forensic departments and judges to work with FIRs, investigations and forensic reports. The system was developed using Hyperledger Fabric with access control features, which makes it closely related to our choice of technology. We also studied NyaYa by Verma et al. (2021), which focuses on electronic legal records and digital evidence. It stores important metadata and hashes on the blockchain while keeping sensitive information and files off-chain using IPFS. This approach is relevant to our project because we also plan to keep sensitive FIR information and documents off-chain while using blockchain for integrity and audit purposes.

We also found a real-world example in Firozabad, India, where the police launched a blockchain-based complaint system in 2022 in collaboration with Polygon. The system allowed citizens to submit complaints, track their status, view the assigned officer and receive updates. This is particularly relevant to our project because it demonstrates that blockchain has been applied in an actual police complaint-management context rather than only being proposed as a research concept. However, the Firozabad system focused primarily on complaint management, whereas our proposed system focuses on the broader FIR lifecycle, investigation workflow, role-based access, post-verification SHA-256 integrity checking and a permissioned Hyperledger Fabric audit layer.

Overall, our research shows that blockchain, hashing and off-chain storage have already been studied for FIRs, police complaints and digital evidence. However, different projects use different technologies, security methods and workflows. Therefore, our project does not claim to be the first blockchain-based FIR system. Instead, our goal is to develop a practical academic prototype for the Pakistan Police context. It combines Hyperledger Fabric, Role-Based Access Control, Investigating Officer verification and SHA-256 hashing after verification. This project will combine these concepts into one simple workflow to demonstrate how blockchain can be used as an additional integrity and audit layer for digital FIR management.

Market Survey:

Existing System	Country	Digital Complaint	Digital FIR/Record	Tracking	Role-Based Access	Blockchain
Punjab Police CMS	Pakistan / Punjab	✓	Related to PSRMS	✓	✓	X
Punjab Police PSRMS	Pakistan / Punjab	-	✓	✓	✓	X
Sindh Police CMS	Pakistan / Sindh	✓	Related	✓	Not fully documented publicly	X
Sindh Police PSRMS	Pakistan / Sindh	-	✓	✓	Not fully documented publicly	X
KP Police digital FIR/record services	Pakistan / KP	✓	✓	✓	Not fully documented publicly	X
Firozabad Police Blockchain Complaint System	India / Firozabad	✓	Related	✓	Not fully documented publicly	✓ Polygon
Proposed Digital FIR	Academic Prototype	✓	✓	✓	✓ RBAC	✓ Hyperledger Fabric

Research Gap Identified from Existing Work

While studying existing FIR systems, we noticed that most of them, such as those used by Punjab, Sindh and KP Police, focus on digital record keeping. They let users register, store, search and track FIRs and complaints but none of the public documentation mentions blockchain or any way to check whether a record was changed later.

We also read research papers on blockchain-based FIR systems. Hassija et al. (2021) propose a consortium blockchain for registering and tracking FIRs. Smart FIR (Khan et al., 2020) uses Ethereum and smart contracts. PoliceChain (Mukherjee & Halder, 2020) uses Hyperledger Fabric with access control for a smart policing system. NyaYa (Verma et al., 2021) keeps case files off-chain in IPFS and stores only the hashes on the blockchain. So blockchain-based FIR systems are not a new idea and we are not claiming they are.

What we found is that each paper uses a different technology and workflow and none of them is designed around the Pakistan Police setting. Our project takes the useful ideas from these papers and combines them in one prototype. It uses:

- A digital FIR workflow with Role-Based Access Control (RBAC)
- Verification by the Investigating Officer
- SHA-256 hashing
- Off-chain storage for sensitive data
- A permissioned Hyperledger Fabric network for audit records

One important choice in our design is when the hash is created. We do not hash a complaint when the citizen submits it, because it may still change during review. The hash is generated only after the Investigating Officer verifies the record or document. This way the blockchain stores a hash of information that has already been checked and any later change can be detected.

Problem Statement:

The main problem is how to maintain the integrity and traceability of digital FIR records and related evidence while protecting sensitive information and controlling access between different users. In a centralized system, the actual FIR data and documents are stored in a database or file storage and the history of changes normally depends on application-level logs and administrative controls. This can make independent verification of important changes difficult. The system therefore needs a way to identify who performed an important action, when the action occurred, what workflow event took place and whether an important document or evidence file was changed after verification. At the same time, sensitive information such as personal details, FIR narratives, investigation notes and documents should not be exposed on a blockchain. The computing problem is therefore to design a hybrid system that combines secure off-chain storage with a permissioned blockchain-based integrity and audit layer.

Proposed Solution:

The proposed project is a Digital FIR Management System that combines a normal secure web-based FIR system with a permissioned Hyperledger Fabric blockchain. The purpose is not to replace existing police systems but to demonstrate how blockchain can be added as an integrity and audit layer to a digital FIR workflow.

The system will use a hybrid architecture. The normal application data will be stored off-chain in PostgreSQL and protected file storage, while selected integrity and audit information will be stored on-chain using Hyperledger Fabric.

Sensitive information such as complainant details, CNIC and contact information, FIR narratives, suspect and witness information, investigation notes and actual documents will not be stored directly on the blockchain. This information will remain in the application database and protected file storage.

The frontend of the system will be developed using React and TypeScript. The backend will be developed using Python and FastAPI, which will handle APIs, authentication, authorization, validation, business logic and communication with the database and blockchain network. PostgreSQL will be used for structured application data.

The system will provide different access levels for users such as citizens, police officers, Investigating Officers, supervisory officers, audit/review users and administrators. Role-Based Access Control (RBAC) will ensure that each user can only perform the operations allowed for their role.

Proposed FIR and Integrity Verification Workflow

The complete proposed workflow will be:

Step 1 - Complaint Submission

The citizen submits a complaint through the web application.

Step 2 - Initial Storage

The submitted information is stored in the application database. At this stage, the system does not create the final blockchain integrity hash.

Step 3 - FIR Review

Authorized police personnel review the complaint and process it according to the defined FIR workflow.

Step 4 - FIR Creation

When the required conditions are satisfied, an FIR is created and assigned a unique FIR identifier.

Step 5 - Officer Assignment

A supervisory officer assigns an Investigating Officer to the FIR.

Step 6 - Investigation

The Investigating Officer records permitted investigation activities, updates and evidence information.

Step 7 - Verification

The Investigating Officer reviews and verifies the relevant FIR information and selected documents.

Step 8 - SHA-256 Hash Generation

After verification, the system generates a SHA-256 hash for the selected verified record or document.

Step 9 - Blockchain Transaction

The hash and required transaction/audit metadata are recorded on Hyperledger Fabric.

Step 10 - Off-Chain Storage

The original FIR information and actual documents remain in PostgreSQL and protected file storage.

Step 11 - Later Integrity Check

When an authorized user requests verification, the system generates a new SHA-256 hash from the current document or record.

Step 12 - Hash Comparison

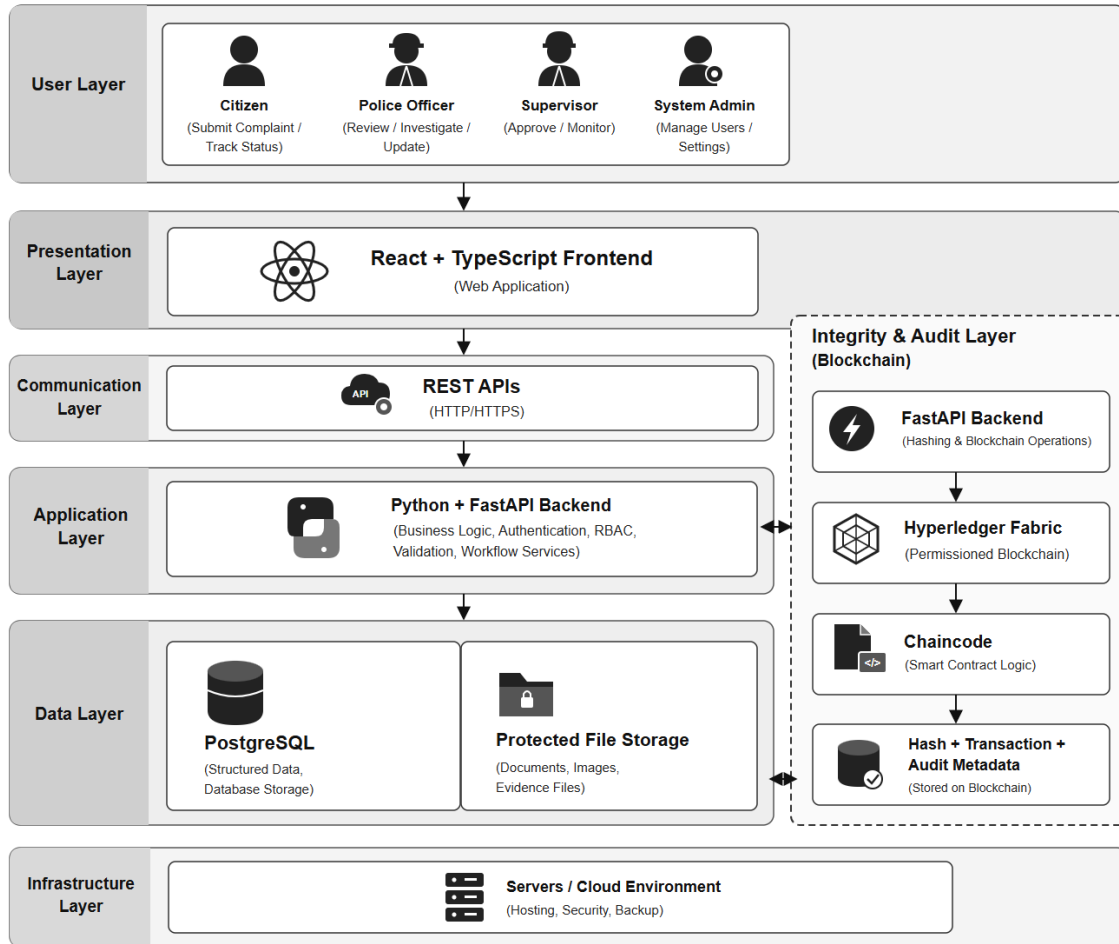
The new hash is compared with the hash previously recorded on the blockchain.

Step 13 - Verification Result

Match: No change was detected.

Mismatch: The system indicates a possible modification after the original verification.

Architecture Diagram



Functional Requirements:

The main functional requirements of the proposed Digital FIR system are:

- **FR-01 - User Login**
The system shall allow registered users to log in using their valid credentials.
- **FR-02 - Role-Based Access:**
The system shall provide different permissions according to the user's role, such as Citizen, Police Officer, Investigating Officer, Supervisory Officer, Audit/Review Officer and Administrator.
- **FR-03 - Complaint Submission:**
The system shall allow citizens to submit complaints through the web application using project/demo data.

- **FR-04 - FIR Creation:**
Authorized police personnel shall be able to review a complaint and create an FIR when the required conditions are satisfied.
- **FR-05 - Unique FIR Number:**
The system shall generate a unique identifier for each registered FIR.
- **FR-06 - FIR Status Management:**
The system shall maintain the status of an FIR throughout its lifecycle, such as Submitted, Under Review, Registered, Assigned, Investigation in Progress, Pending Review, Closed and Rejected.
- **FR-07 - Officer Assignment:**
An authorized supervisory officer shall be able to assign an Investigating Officer to a registered FIR.
- **FR-08 - Investigation Updates:**
The Investigating Officer shall be able to record permitted investigation activities and case updates.
- **FR-09 - Evidence Management:**
Authorized users shall be able to register evidence information and upload permitted test documents.
- **FR-10 - Document Hashing:**
After the required verification, the system shall generate a SHA-256 hash for selected FIR records or documents.
- **FR-11 - Blockchain Recording:**
The system shall record selected hashes and required audit information on the Hyperledger Fabric blockchain.
- **FR-12 - Integrity Verification:**
The system shall generate a new hash when an authorized user requests verification and compare it with the previously recorded blockchain hash.
- **FR-13 - Audit Trail:**
The system shall maintain a history of important FIR activities, including the user, action and time of the activity.
- **FR-14 - Search and Filtering:**
Authorized users shall be able to search and filter FIR records using permitted information such as FIR number, date, status, station and assigned officer.
- **FR-15 - Notifications:**
The system shall provide notifications for important workflow events, such as FIR assignment or status changes.
- **FR-16 - Reports and Dashboard:**
The system shall provide role-specific dashboards and basic reports for authorized users.

- **FR-17 - User Management:**

The administrator shall be able to manage users, roles and selected system settings.

Non-Functional Requirements:

The non-functional requirements describe how the system should work and the quality expected from the system.

- **NFR-01 - Security:**

The system shall protect user accounts, FIR information and uploaded documents from unauthorized access.

- **NFR-02 - Confidentiality:**

Sensitive FIR information shall not be stored directly on the blockchain and shall only be accessible to authorized users.

- **NFR-03 - Integrity:**

The system shall use SHA-256 hashing and blockchain records to detect changes in selected verified documents or records.

- **NFR-04 - Performance:**

The system should provide reasonable response times for normal operations during the expected project workload.

- **NFR-05 - Availability:**

The system should remain available during normal operation and handle common failures properly.

- **NFR-06 - Usability:**

The interface should be simple, understandable and easy to use for different types of users.

- **NFR-07 - Maintainability:**

The system shall use separate and organized frontend, backend, database and blockchain components so that the project can be maintained easily.

- **NFR-08 - Scalability:**

The system architecture should allow additional police stations or organizations to be added in the future.

- **NFR-09 - Reliability:**

The system shall handle invalid input, unauthorized requests and failed operations without causing incorrect data to be stored.

- **NFR-10 - Compatibility:**

The web application should work on commonly used modern web browsers.

- **NFR-11 - Auditability:**

Important activities shall be recorded with the responsible user and timestamp so that the activity can be reviewed later.

- **NFR-12 - Data Privacy:**
Personal information and sensitive case information shall be kept off-chain unless there is a specific reason to store a non-sensitive reference or hash.

Security Requirements:

Security is an important part of the proposed system because FIR records may contain sensitive personal and case-related information. The following security requirements will be considered during development:

- **SEC-01 - Authentication:**
Only registered users shall be allowed to access protected parts of the system.
- **SEC-02 - Password Protection:**
User passwords shall be stored using a secure password-hashing method such as Argon2id or bcrypt. Plain-text passwords shall not be stored.
- **SEC-03 - Role-Based Authorization:**
The backend shall check the user's role before allowing access to protected operations.
- **SEC-04 - Secure Tokens:**
Authentication tokens shall have an appropriate expiry time and shall be handled securely.
- **SEC-05 - Input Validation:**
The backend shall validate user input before processing or storing it.
- **SEC-06 - File Security:**
Uploaded files shall be checked for allowed file types and sizes. Files shall not be directly exposed through unrestricted public URLs.
- **SEC-07 - Database Security:**
The application shall use secure database access methods to reduce risks such as SQL injection.
- **SEC-08 - Access Control:**
Users shall only be able to view or modify FIR information that their role allows them to access.
- **SEC-09 - Audit Logging:**
Important security and system activities shall be recorded with the user, action and timestamp.
- **SEC-10 - Data Protection:**
Sensitive information shall remain in protected off-chain storage and shall not be unnecessarily stored on the blockchain.
- **SEC-11 - Blockchain Access Control:**
Only authorized identities and system components shall be allowed to submit transactions to the permissioned Hyperledger Fabric network.

- **SEC-12 - Integrity Verification:**
The system shall use SHA-256 hashes and blockchain records to identify possible changes to previously verified records or documents.
- **SEC-13 - Secure Communication:**
Where the system is deployed over a network, HTTPS/TLS shall be used to protect communication between the client and server.
- **SEC-14 - Error Handling:**
The system shall not expose passwords, tokens, database information or other sensitive technical details in error messages.

Scope of the Project:

Module 1 - Authentication and Role Management

The system will provide secure login and authentication for different types of users. Role-Based Access Control will be used to assign permissions according to the user's role. The main roles will include Citizen, Police Officer, Investigating Officer, Supervisory Officer, Audit/Review user and Administrator. The backend will check authorization before protected operations are performed. Access logs will also be maintained for important user activities. This module will ensure that users can only access the functions and information allowed for their role.

Module 2 - Complaint and FIR Management

Citizens will be able to submit complaint information through the web interface using synthetic project data. Authorized police users will review the complaint and create the FIR when the required conditions are satisfied. The system will assign a unique FIR identifier and maintain the FIR status throughout its lifecycle. The workflow will include stages such as Submitted, Under Review, Registered, Assigned, Investigation in Progress, Pending Review, Closed or Rejected. Users will only be allowed to perform status changes according to their role and permissions.

Module 3 - Investigation and Officer Assignment

Supervisory users will be able to assign an Investigating Officer to a registered FIR. The Investigating Officer will record permitted investigation events and case updates through the system. Each important action will be associated with the responsible user and timestamp. The system will allow authorized users to view the progress of an FIR according to their role. This module will provide a clear history of investigation activities. The information will remain stored off-chain in the application database.

Module 4 - Evidence Management and SHA-256 Integrity Verification

Authorized users will be able to register evidence metadata and upload relevant test documents. The actual documents will remain in protected off-chain file storage. The system will not apply the final integrity hash at the initial submission stage. First, the application and its relevant documents will be reviewed and verified by the Investigating Officer. After verification, the system will generate a SHA-256 hash for the selected verified document or record. Only the hash and required metadata will be stored on Hyperledger Fabric. During later verification, the system will generate a new hash and compare it with the blockchain hash to detect possible modification.

Module 5 - Hyperledger Fabric and Audit Trail

The project will use Hyperledger Fabric as a permissioned blockchain network. Only authorized participants and system components will be allowed to submit or validate blockchain transactions. Chaincode will define the required blockchain operations and workflow rules. The blockchain will store selected hashes, transaction identifiers, workflow events, timestamps, status information and required audit metadata instead of sensitive FIR information. This will provide a tamper-evident history of important verified activities. The blockchain will act as an integrity and audit layer and will not replace PostgreSQL as the main application database.

Module 6 - Dashboard, Search, Notifications and Administration

The system will provide role-specific dashboards for authorized users. Users will be able to search and filter FIR records according to permitted fields such as FIR number, date, status, station and assigned officer. Supervisory users will be able to view case progress and relevant reports. The system will provide notifications for important workflow events where required. Administrators will manage users, roles and system-level configuration. These features will help users monitor the FIR workflow and access information according to their permissions.

List of Faculty Proposed Changes

Digital FIR: A Blockchain-Based Secure First Information Report Management System

Supervisor's Signature: _____

Proposed Change	Proposed By	Supervisor's Decision
Lack of background knowledge and limited literature review. Please study deeply and then apply concepts	Mubariz Rehman	
methodology is well defined complexity need to enhance in project	Hassan Siddique	
You need to go deep down to explore and get clarity what will be your project output. Work on suggestions	Tajamul Shahzad	
Conduct an in-depth study of the existing FIR system/process to identify its workflow, limitations and opportunities for improvement.	Zeeshan Ali	
+ Market survey of similar software applications is recommended + Literature review of related work is recommended + Domain knowledge of the students is weak	Sajid Qureshi	
Domain knowledge problem.	Sharjeel Gillani	
Fine	Abdul Mateen	

Project Plan

- 1. Project Management**
 - 1.1 Proposal and Feedback Management**
 - 1.2 WBS and Responsibility Matrix**
 - 1.3 GitHub and Version Control**
 - 1.4 Progress and Change Tracking**
- 2. Research and Domain Study**
 - 2.1 Literature Review**
 - 2.2 FIR Domain and Workflow Study**
 - 2.3 Existing FIR Systems and Application Survey**
 - 2.4 Requirements and Gap Analysis**
- 3. System Design**
 - 3.1 System Design (roles, use cases and permissions)**
 - 3.2 Workflow Design (FIR states and verification)**
 - 3.3 Architecture Design**
 - 3.4 Database Design**
 - 3.5 Blockchain Design**
 - 3.6 Hashing Design**
- 4. Implementation**
 - 4.1 Frontend**
 - 4.2 Backend**
 - 4.3 FIR Modules**
 - 4.4 Data Management**
 - 4.5 Verification**
 - 4.6 Integrity**
 - 4.7 Blockchain**
 - 4.8 Chaincode and Audit Transactions**
- 5. Testing and Evaluation**
 - 5.1 Functional Testing**
 - 5.2 Authentication and RBAC Testing**
 - 5.3 Integrity Testing**
 - 5.4 Blockchain Transaction Testing**
 - 5.5 API & File Security Testing**
 - 5.6 Performance Evaluation**
- 6. Documentation and Delivery**
 - 6.1 User & Administration Documentation**
 - 6.2 Final Project Report**
 - 6.3 Deployment Package**
 - 6.4 Final Presentation and Viva Preparation**

Roles & Responsibility Matrix:

The matrix identifies responsibility for each major deliverable. Research, documentation, integration, code review, security testing, final report, presentation and viva preparation are shared across all three members.

WBS #	WBS Deliverable	Activity #	Activity to Complete the Deliverable	Duration (Days)	Responsible Team Members & Roles
1	Project Management	1.1	Update proposal, track feedback and maintain change control	8	Sharaiz Ahmed, Architecture and Blockchain Lead; All members, shared project work
2	Literature Review	2.1	Study NyaYa, Smart FIR, PoliceChain and related blockchain-FIR research	12	All members
2	FIR Domain Study	2.2	Study FIR workflow, roles and verification process	8	All members
2	Existing System Study	2.3	Study Punjab, Sindh, KP and related FIR applications	10	All members
2	Requirements	2.4	Define functional/non-functional requirements and gap	8	All members
3	System Design	3.1	Define roles, use cases and permissions	7	Sharaiz Ahmed, Architecture and Blockchain Lead; Abdul Moiz, Backend and Database Lead
3	Workflow Design	3.2	Define FIR states and verification workflow	7	Sharaiz Ahmed, Architecture and Blockchain Lead; Basit Ali, Frontend and Testing Lead
3	Architecture	3.3	Design hybrid off-chain/on-chain architecture	10	Sharaiz Ahmed, Architecture and Blockchain Lead

3	Database Design	3.4	Design PostgreSQL schema and file storage	8	Abdul Moiz, Backend and Database Lead
3	Blockchain Design	3.5	Design Fabric network and chaincode	12	Sharaiz Ahmed, Architecture and Blockchain Lead
3	Hashing Design	3.6	Define post-verification SHA-256 workflow	5	Sharaiz Ahmed, Architecture and Blockchain Lead; Abdul Moiz, Backend and Database Lead
4	Frontend	4.1	Develop React/TypeScript pages	20	Basit Ali, Frontend and Testing Lead
4	Backend	4.2	Develop FastAPI APIs and RBAC	25	Abdul Moiz, Backend and Database Lead
4	FIR Modules	4.3	Develop complaint, FIR and investigation modules	20	Abdul Moiz, Backend and Database Lead; Basit Ali, Frontend and Testing Lead
4	Data Management	4.4	PostgreSQL and protected file storage	12	Abdul Moiz, Backend and Database Lead
4	Verification	4.5	Implement Investigating Officer verification	8	Abdul Moiz, Backend and Database Lead; Sharaiz Ahmed, Architecture and Blockchain Lead
4	Integrity	4.6	Implement SHA-256 generation and verification	10	Sharaiz Ahmed, Architecture and Blockchain Lead; Abdul Moiz, Backend and Database Lead
4	Blockchain	4.7	Integrate Hyperledger Fabric	20	Sharaiz Ahmed, Architecture

					and Blockchain Lead; Abdul Moiz, Backend and Database Lead
4	Chaincode	4.8	Implement blockchain transactions and audit metadata	15	Sharaiz Ahmed, Architecture and Blockchain Lead
5	Functional Testing	5.1	Test complete FIR workflow	8	Basit Ali, Testing Lead; All members
5	Security Testing	5.2	Test authentication/RBAC/authorization	7	Abdul Moiz, Backend and Database Lead; Sharaiz Ahmed, Architecture and Blockchain Lead
5	Integrity Testing	5.3	Modify test document and verify hash mismatch detection	7	Sharaiz Ahmed, Architecture and Blockchain Lead; Basit Ali, Testing Lead
5	Blockchain Testing	5.4	Test transactions and audit history	7	Sharaiz Ahmed, Architecture and Blockchain Lead
5	Security Testing	5.5	API/file/input security checks	7	Abdul Moiz, Backend and Database Lead
5	Performance	5.6	Basic API/blockchain performance evaluation	7	All members
6	Documentation	6.1	User and administration documentation	10	All members
6	Final Report	6.2	Prepare final report	10	All members
6	Deployment Package	6.3	Prepare demonstration environment	7	Sharaiz Ahmed, Architecture and Blockchain Lead (GitHub and deployment coordination)

6	Presentation	6.4	Presentation and viva preparation	7	All members
---	--------------	-----	-----------------------------------	---	-------------

Approval

Project Supervisor

Comments _____

Name: _____

Date: _____ Signature: _____

Project Coordinator

Comments _____

Name: _____

Date: _____ Signature: _____